curve in Figure 30.5. The signal travels forward in local, bulk time, but backward in bedroom time, arriving before it started out.[57] It is this gravitational signal that pushes the book out of the bookcase and twitches the watch's second hand.

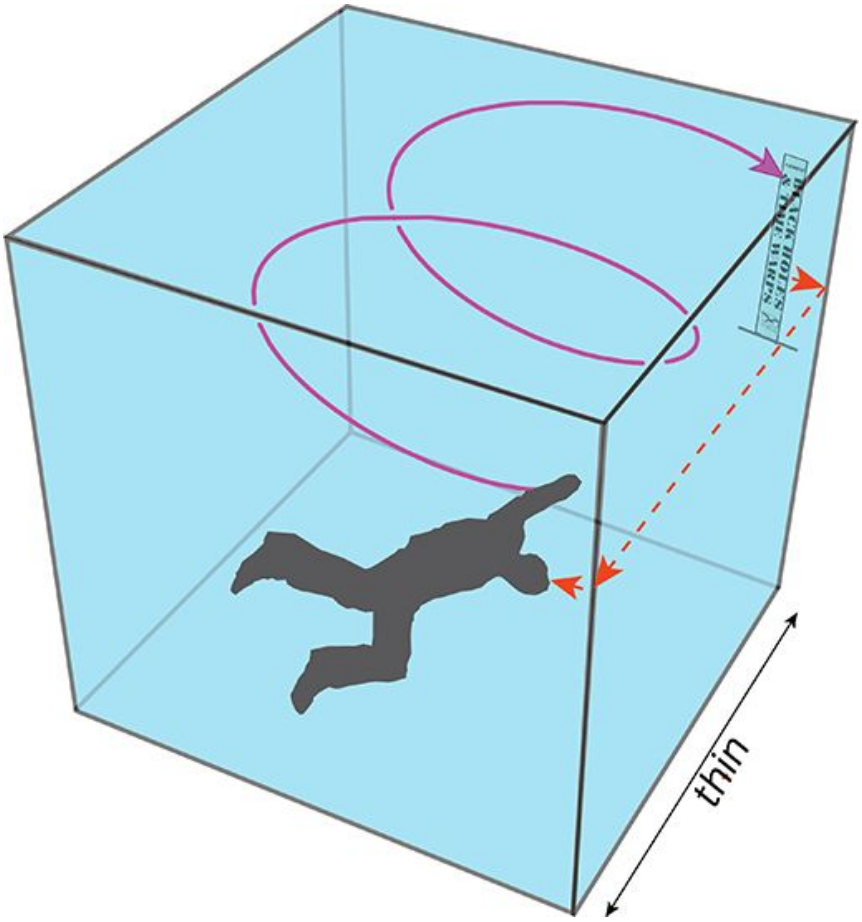


Fig. 30.5. A Cooper icon sees a book via the red dashed light ray and exerts a force on the book via a gravitational signal that spirals along the violet curve. I've suppressed one of our brane's spatial dimensions.

This is rather like one of my favorite Escher drawings, *Waterfall* (Figure 30.6). Downward in the drawing is analogous to the forward flow of bedroom time, and the flowing water is analogous to the forward flow of local time. A leaf on the water is carried forward with the water just like signals in the bulk are carried forward in local time.